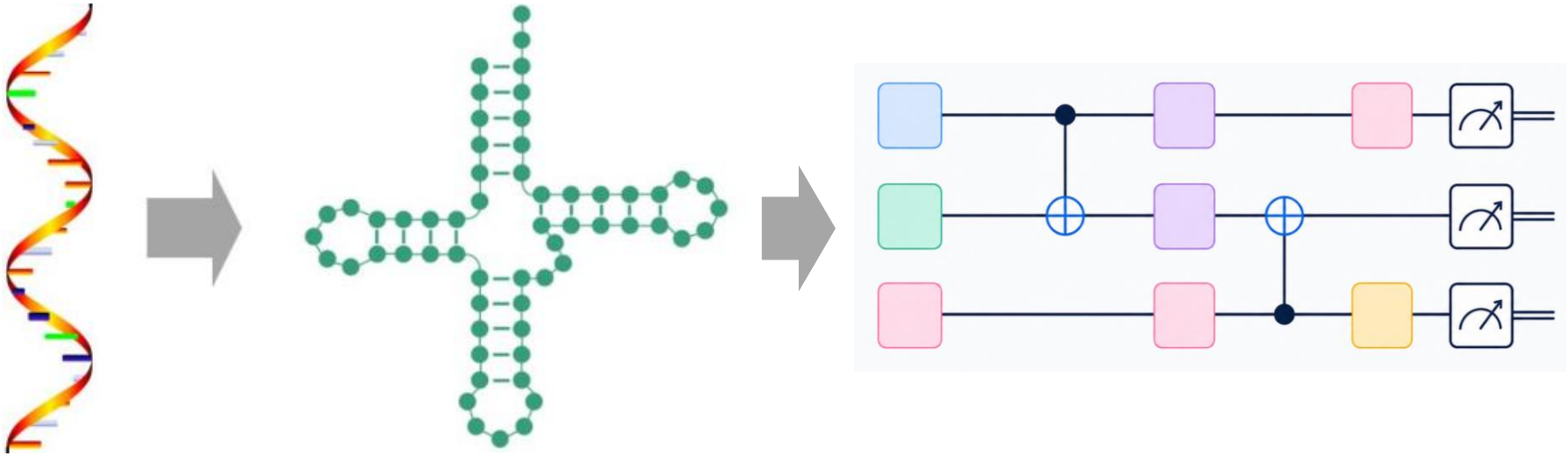
Where Should RNA Constraints Live?

Testing strict and hybrid QUBO formulations for RNA secondary structure

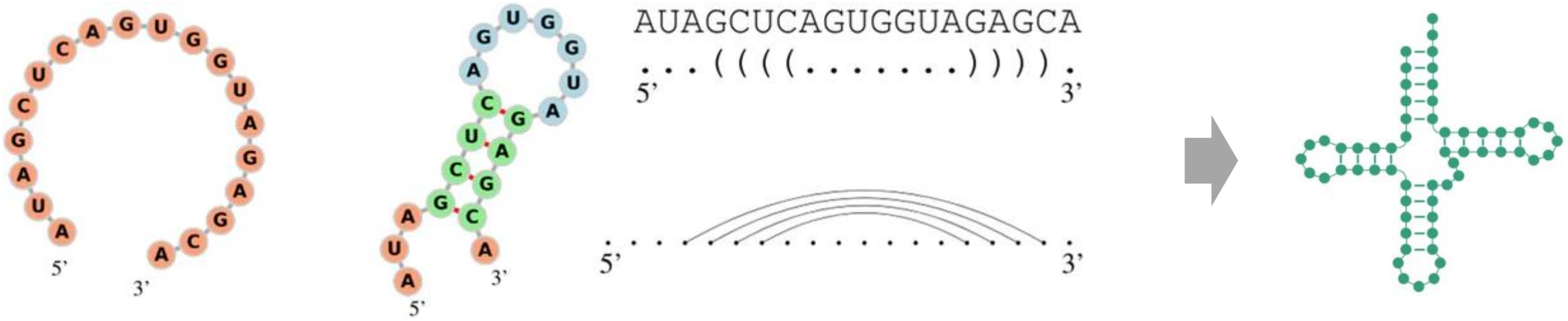


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figure adapted from: Zhao, Q., Zhao, Z., Fan, X., Yuan, Z., Mao, Q., & Yao, Y. (2020). Review of machine learning methods for RNA secondary structure prediction

Project question

Where should structural constraints be enforced:
In the quantum objective or during classical postprocessing?



- mRNA can fold when complementary nucleotides form base pairs
- The resulting secondary structure can affect stability, translation, and manufacturing
- Classical tools such as ViennaRNA predict structures using thermodynamic models
- Quantum optimization offers another way to explore combinations of candidate RNA stems

Figure adapted from: Marchei, D., & Merelli, E. (2022). RNA secondary structure factorization in prime tangles. BMC bioinformatics, 23(Suppl 6), 345.

Three QUBO formulations

Shared sequences, candidate stems, quantum workflow, repair, and ViennaRNA evaluation
The difference is **where** conflicts are resolved

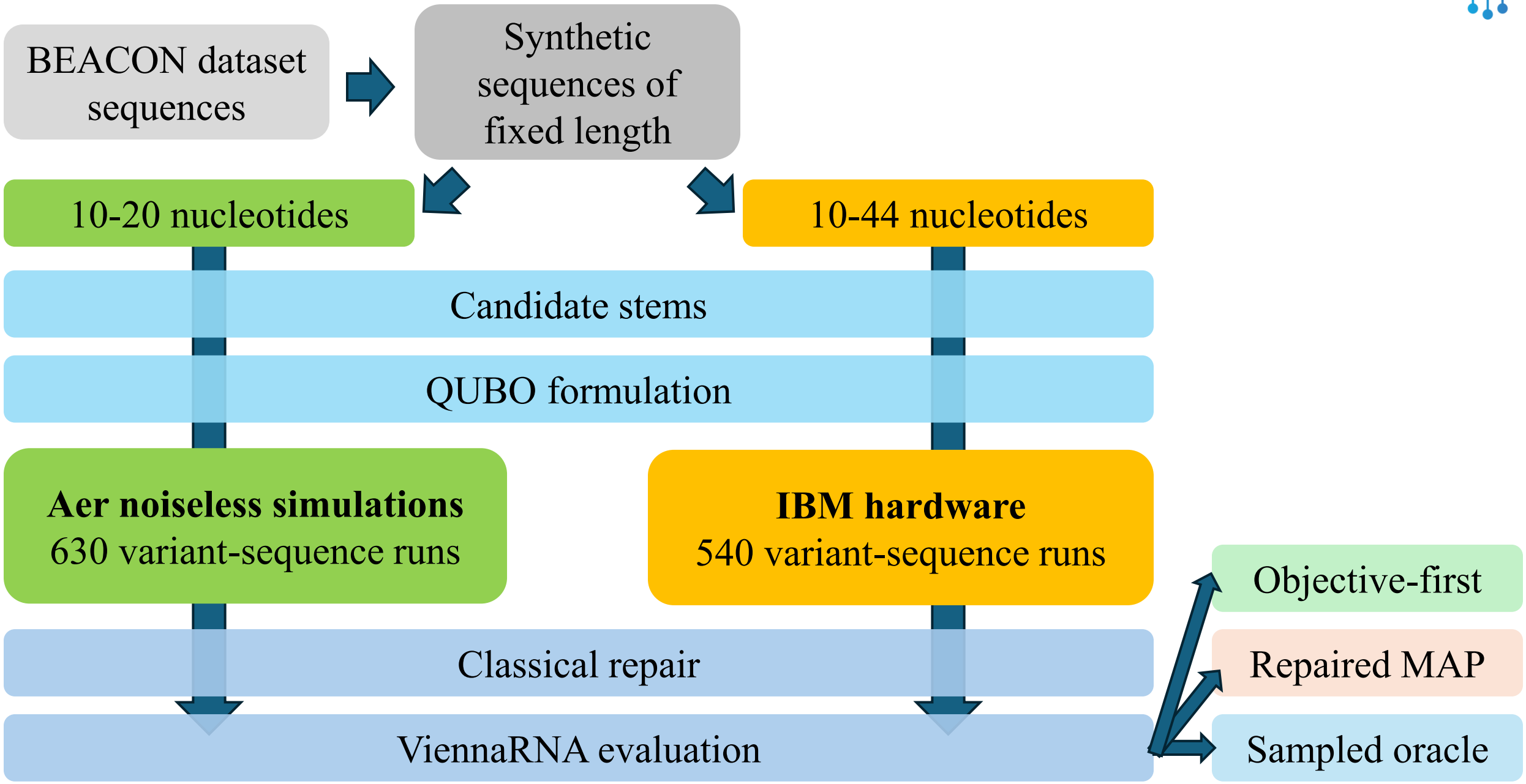
more quantum constraints ⟵ ⟶ more classical repair

Strict	Relaxed	Postprocessed
Penalizes nucleotide overlap	Penalizes nucleotide overlap	Overlaps repaired classically
Penalizes crossing stems	Crossing stems repaired after sampling	Crossing stems repaired after sampling
Constraints handled inside the QUBO	Intermediate quantum complexity	Simplest quantum objective

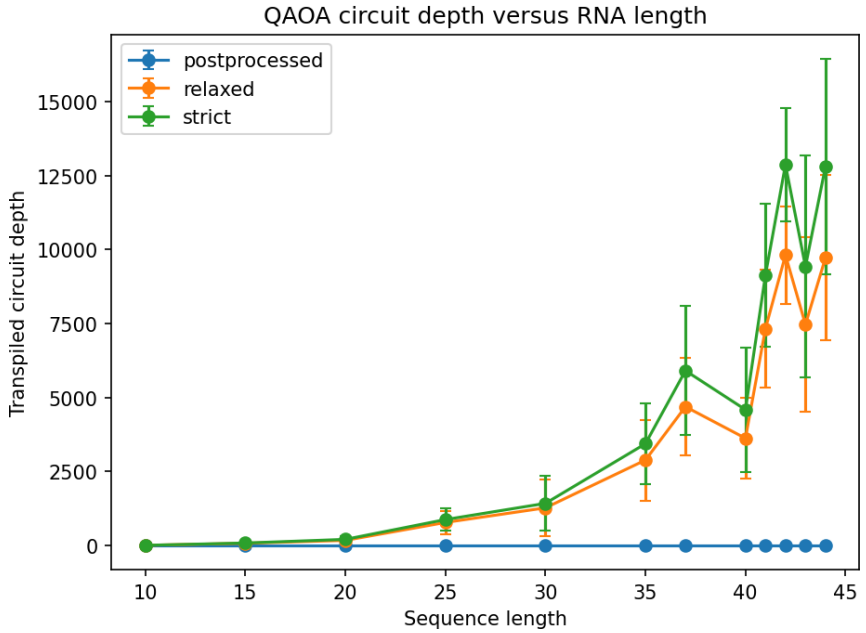
Modeling choice:
Instead of embedding full thermodynamic MFE, this project uses candidate-stem length as a simplified structural proxy. ViennaRNA is used after sampling as an independent benchmark

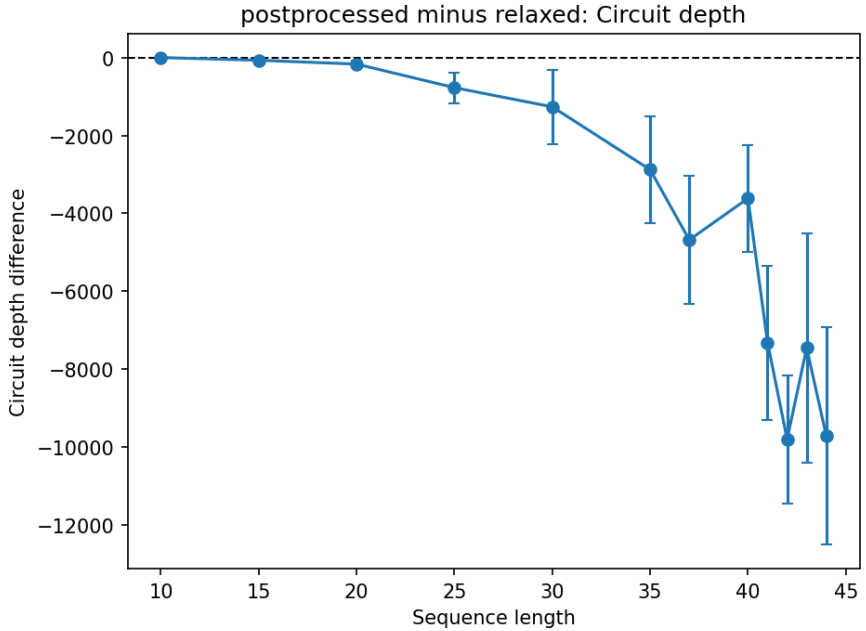
From RNA Sequences to Quantum Measurements





Results - Fewer Constraints, Shallower Circuits





- compared with strict encoding, the relaxed variant used **37% fewer** quadratic interactions on average

- mean circuit **depth decreased by 21.3%**

- relaxed circuits were **shallower** at every tested nontrivial sequence length

- the sampled-oracle quality remained very **similar**:

Metric	Relaxed	Strict
Energy gap	0.443	0.475
Base-pair F1	0.801	0.811

The crossing penalties added noticeable quantum cost but did not substantially improve the best repaired candidates that were sampled

Results - Good candidates are easier to find than to select

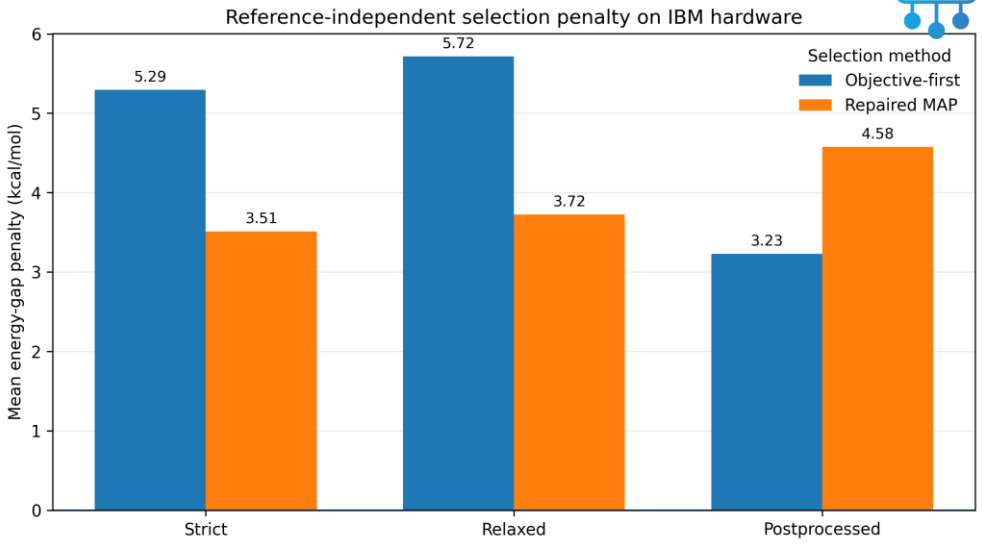
- the sampled oracle shows that **strong repaired candidates** were **present** in the measured samples

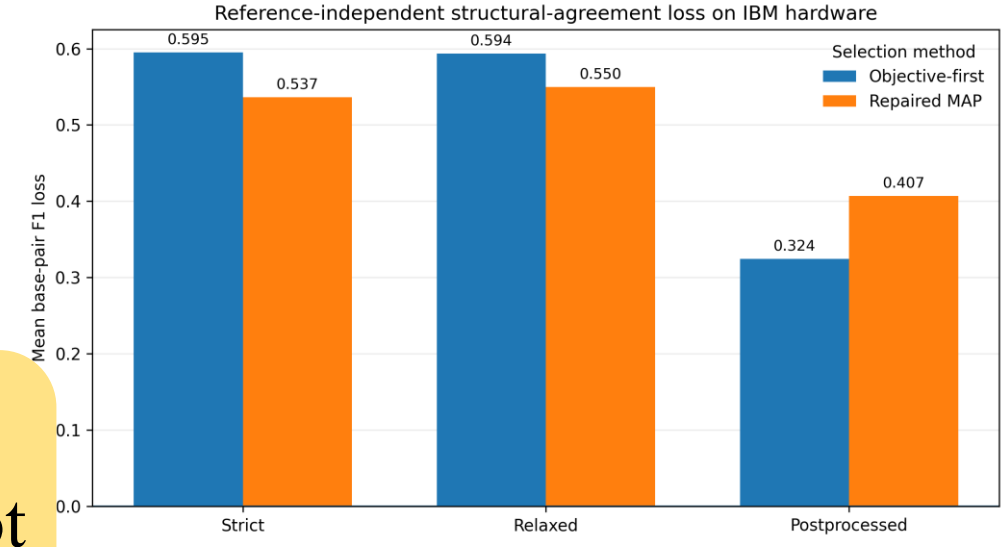
- reference-independent selection produced **weaker** results

- on IBM hardware, objective-first increased the mean energy-gap penalty by 5.3 kcal/mol for strict and by 5.7 kcal/mol for relaxed

- repaired MAP reduced penalties to 3.5 and 3.7 kcal/mol

- base-pair F1 losses also improved with repaired MAP





Strict and relaxed QUBOs sampled useful structures, but the QUBO objective alone was not sufficient to identify the best biological candidate

Results - Trade-off summary

- Strict variant improved raw validity mainly for smaller instances
- From sequence length 35 onward, classical repair became essential for all variants
- Postprocessed variant kept circuit depth low but shifted all the work to repair

more quantum constraints → more classical repair

Strict	Relaxed	Postprocessed
Highest constraint cost	Best overall balance	Lowest circuit cost
Better small-case validity	Lower depth, similar candidate potential	Highest repair burden
Limited scaling benefit	Best tested compromise	Weak repaired outcomes

Conclusion, Limitations, Future Work

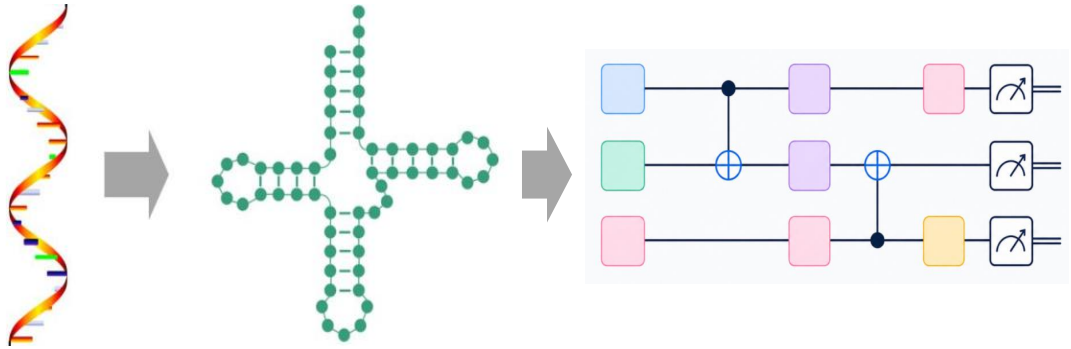
Among the tested strategies, **relaxed encoding** gave the **best balance** between quantum resource cost, classical repair burden, and sampled candidate quality

Selected limitations

- The QUBO rewards stem length, not a full thermodynamic free energy
- Sampled oracle is an upper bound, not native prediction accuracy
- Results use shallow p=1 QAOA, shorter sequences, one hardware backend

Future work

- Prioritizing reference-independent selection
- Better thermodynamic model rewards
- Hardware-aware optimization



Careful placement of constraints may matter as much as the quantum optimizer

Where Should RNA Constraints Live?



Thank you
for your attention



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